

GRADE 4 MATHEMATICS

Workbook Answer Key: Subtraction & Its Properties

This document provides the complete, mathematically verified answer key and step-by-step solutions for all exercises in the **Grade 4 Subtraction and Its Properties** chapter. It is designed for teachers, tutors, and parents to easily review students' work and guide them through mathematical problem-solving.

Tune Up (Page 24)

Question: Meena went to market to buy some clothes. She had ₹ 1500 in her purse. She met a friend, Ajay, who was also in the market to buy a pair of skating shoes. Ajay told her that he had ₹ 300 less to buy the skating shoes. Meena offered him to borrow ₹ 300 from her to buy the skating shoes.

How much money Meena has after giving ₹ 300 to Ajay?

Solution:

- Total money Meena had initially = ₹ 1500
- Money given to Ajay = ₹ 300
- Money left with Meena = ₹ 1500 - ₹ 300 = ₹ 1200

Exercise 3A (Pages 36-37)

1. Arrange the following numbers in columns and subtract them:

Q	Problem	Answer & Breakdown	Q	Problem	Answer & Breakdown
a	7654 - 3243	4,411 (7654 - 3243)	g	43,791 - 23,448	20,343 (43,791 - 23,448)
b	97,205 - 3426	93,779 (97,205 - 3426)	h	50,408 - 3412	46,996 (50,408 - 3412)
c	28,976 - 1167	27,809 (28,976 - 1167)	i	7,56,320 - 3,47,980	4,08,340 (7,56,320 - 3,47,980)
d	45,780 - 34,271	11,509 (45,780 - 34,271)	j	9,32,447 - 8,76,125	56,322 (9,32,447 - 8,76,125)
e	26,754 - 23,842	2,912 (26,754 - 23,842)	k	78,654 - 70,430	8,224 (78,654 - 70,430)
f	87,646 - 6524	81,122 (87,646 - 6524)	l	8,74,506 - 4,01,228	4,73,278 (8,74,506 - 4,01,228)

2. Solve the following:

Q	Problem	Answer	Q	Problem	Answer
a	Subtract 20,365 from 78,654	$78,654 - 20,365 =$ 58,289	f	Subtract 475 from 56,743	$56,743 - 475 =$ 56,268
b	Diff between 56,435 and 8,76,540	$8,76,540 - 56,435 =$ 8,20,105	g	Subtract 8576 from 85,765	$85,765 - 8,576 =$ 77,189
c	Subtract 3,45,270 from 7,86,775	$7,86,775 - 3,45,270 =$ 4,41,505	h	Diff between 9,87,659 and 76,456	$9,87,659 - 76,456 =$ 9,11,203
d	Diff between 4,57,869 and 7865	$4,57,869 - 7,865 =$ 4,50,004	i	Subtract 9870 from 90,000	$90,000 - 9,870 =$ 80,130
e	Subtract 45,636 from 78,670	$78,670 - 45,636 =$ 33,034	j	Diff between 75,683 and 75,863	$75,863 - 75,683 =$ 180

3. Find the difference between six lakh forty-three thousand three hundred seven and six lakh four thousand two hundred seven.

- First number = 6,43,307
- Second number = 6,04,207
- Difference: $6,43,307 - 6,04,207 =$ **39,100**

4. By how much is 12,458 bigger than 10,245?

- Comparison: $12,458 - 10,245 =$ **2,213**. (It is 2,213 bigger)

5. By how much is the sum of 1,51,263 and 47,856 smaller than 2,00,000?

- Sum of numbers = $1,51,263 + 47,856 =$ 1,99,119
- Comparison with 2,00,000 = $2,00,000 - 1,99,119 =$ **881**. (It is 881 smaller)

Exercise 3B (Page 38)

1. Fill in the blanks using the properties of subtraction:

a) $47,690 - 1 = 47,689$	g) $61,894 - 18,221 = 43,673$
b) $73,000 - 1 = 72,999$	h) $79,653 - 1 = 79,652$
c) $9,53,345 - 0 = 9,53,345$	i) $5,50,000 - 1 = 5,49,999$
d) $4,33,210 - 4,33,210 = 0$	j) $54,635 - 33,101 = 21,534$
e) $57,290 - 57,290 = 0$	k) $7,56,398 - 4,10,728 = 3,45,670$
f) $43,673 - 18,221 = 25,452$	l) $8,87,583 - 8,82,926 = 4,657$

2. What number should be subtracted from 7,59,593 to get a difference of 23,943?

- Let the number be X. Then, $7,59,593 - X = 23,943$
- $X = 7,59,593 - 23,943 = 7,35,650$

3. When we subtract 35,495 from a number, we get a difference of 67,435. Find out the number.

- Let the number be Minuend. Minuend - Subtrahend = Difference
- Minuend - 35,495 = 67,435
- Minuend = $67,435 + 35,495 = 1,02,930$

Apply It (Page 39) & Exercise 3C (Pages 39-40)

Apply It (Page 39): Capacity of Eden Gardens stadium is 67,507. During a cricket match, if 34,612 people have entered before noon and 14,708 between noon and 2 o'clock, find how many more people can enter the field till it is filled to capacity.

- Total people entered = $34,612 + 14,708 = 49,320$
- Remaining capacity = $67,507 - 49,320 = 18,187$ people

Exercise 3C Word Problems:

1. Sushma ordered 15,775 yellow and white flowers... 9,455 yellow flowers, rest white.

- Total flowers = 15,775
- Yellow flowers = 9,455
- White flowers = $15,775 - 9,455 = 6,320$ white flowers

2. There were 2,255 seats in an indoor stadium. 1,975 people came... vacant seats?

- Total seats = 2,255
- Attending people = 1,975
- Vacant seats = $2,255 - 1,975 = 280$ seats

3. Library has 3,975 books. Grade 8 issued 120 books, Grade 9 issued 95 books. Books left?

- Total books issued = $120 + 95 = 215$
- Books left in library = $3,975 - 215 = \mathbf{3,760 \text{ books}}$

4. Total of 5,075 sheets of paper were used from a pile of 9,600 sheets. Sheets left?

- Initial sheets = 9,600
- Used sheets = 5,075
- Sheets left = $9,600 - 5,075 = \mathbf{4,525 \text{ sheets}}$

5. 1,344 cans of health drinks. 1,275 distributed. Cans left unused?

- Total cans = 1,344
- Distributed = 1,275
- Unused left = $1,344 - 1,275 = \mathbf{69 \text{ cans}}$

6. 5,360 white tiles and 2,225 blue tiles bought. 399 left unused. How many tiles used?

- Total tiles bought = $5,360 + 2,225 = 7,585$
- Unused tiles = 399
- Tiles used = $7,585 - 399 = \mathbf{7,186 \text{ tiles}}$

7. Shopkeeper ordered 25,000 boxes. Had 1,997 unsold last year. Left with 87 boxes now. Boxes sold this year?

- Total stock = $25,000 \text{ (ordered)} + 1,997 \text{ (from last year)} = 26,997$
- Unsold remaining = 87
- Boxes sold this year = $26,997 - 87 = \mathbf{26,910 \text{ boxes}}$

Exercise 3D (Page 41)

1. Subtract and verify the answers using addition:

Q	Problem	Difference	Verification (Diff + Subtrahend = Minuend)
a	$5,46,389 - 87,598$	4,58,791	$4,58,791 + 87,598 = \mathbf{5,46,389}$ (Verified)
b	$75,683 - 8,457$	67,226	$67,226 + 8,457 = \mathbf{75,683}$ (Verified)
c	$4,92,022 - 98,594$	3,93,428	$3,93,428 + 98,594 = \mathbf{4,92,022}$ (Verified)
d	$85,493 - 9,485$	76,008	$76,008 + 9,485 = \mathbf{85,493}$ (Verified)

2. Solve the following and verify the answers using addition:

Q	Problem	Difference	Verification (Diff + Subtrahend = Minuend)
a	Subtract 46,573 from 5,76,937	$5,76,937 - 46,573 = \mathbf{5,30,364}$	$5,30,364 + 46,573 = \mathbf{5,76,937}$ (Verified)
b	Subtract 8,478 from 94,579	$94,579 - 8,478 = \mathbf{86,101}$	$86,101 + 8,478 = \mathbf{94,579}$ (Verified)
c	Subtract 858 from 4,57,635	$4,57,635 - 858 = \mathbf{4,56,777}$	$4,56,777 + 858 = \mathbf{4,57,635}$ (Verified)
d	Subtract 48,541 from 5,76,836	$5,76,836 - 48,541 = \mathbf{5,28,295}$	$5,28,295 + 48,541 = \mathbf{5,76,836}$ (Verified)

Exercise 3E (Page 42)

1. Estimate the difference to the nearest Ten and compare it with the actual difference:

Problem	Rounding to Nearest Ten	Estimated Diff	Actual Diff	Comparison
a) $49 - 23$	$49 \rightarrow 50, 23 \rightarrow 20$	$50 - 20 = \mathbf{30}$	$49 - 23 = \mathbf{26}$	30 is close to 26
b) $76 - 54$	$76 \rightarrow 80, 54 \rightarrow 50$	$80 - 50 = \mathbf{30}$	$76 - 54 = \mathbf{22}$	30 is close to 22
c) $177 - 123$	$177 \rightarrow 180, 123 \rightarrow 120$	$180 - 120 = \mathbf{60}$	$177 - 123 = \mathbf{54}$	60 is close to 54
d) $8,795 - 6,741$	$8,795 \rightarrow 8,800, 6,741 \rightarrow 6,740$	$8,800 - 6,740 = \mathbf{2,060}$	$8,795 - 6,741 = \mathbf{2,054}$	2,060 is close to 2,054

2. Estimate the difference to the nearest Hundred and compare it with the actual difference:

Problem	Rounding to Nearest Hundred	Estimated Diff	Actual Diff	Comparison
a) $287 - 122$	$287 \rightarrow 300, 122 \rightarrow 100$	$300 - 100 = \mathbf{200}$	$287 - 122 = \mathbf{165}$	200 is close to 165
b) $76,840 - 43,656$	$76,840 \rightarrow 76,800, 43,656 \rightarrow 43,700$	$76,800 - 43,700 = \mathbf{33,100}$	$76,840 - 43,656 = \mathbf{33,184}$	33,100 is close to 33,184
c) $4,56,894 - 3,54,536$	$4,56,894 \rightarrow 4,56,900, 3,54,536 \rightarrow 3,54,500$	$4,56,900 - 3,54,500 = \mathbf{1,02,400}$	$4,56,894 - 3,54,536 = \mathbf{1,02,358}$	1,02,400 is close to 1,02,358
d) $2333 - 1290$	$2,333 \rightarrow 2,300, 1,290 \rightarrow 1,300$	$2,300 - 1,300 = \mathbf{1,000}$	$2,333 - 1,290 = \mathbf{1,043}$	1,000 is close to 1,043

3. At a book fair, 8643 books were sold on Saturday and Sunday. If 4324 books were sold on Saturday, estimate the number of books sold on Sunday.

Rounding to the nearest Thousand (recommended for larger numbers at Grade 4 level):

- Total books sold $\approx 9,000$ (8,643 rounded)
- Saturday books sold $\approx 4,000$ (4,324 rounded)
- Estimated Sunday books sold $= 9,000 - 4,000 = \mathbf{5,000 \text{ books}}$

Alternative (rounding to nearest Hundred): $8,600 - 4,300 = \mathbf{4,300 \text{ books}}$. (Actual Sunday books sold $= 8,643 - 4,324 = 4,319$).

Workout (Pages 42-43)

1. Solve the following:

a) $7,54,289 - 87,355 = \mathbf{6,66,934}$	d) $1,25,389 - 4,882 - 34,703 = \mathbf{85,804}$
b) $5,80,156 - \mathbf{4,03,598} = 1,76,558$	e) $9,58,674 - (8,675 + 24,874) = \mathbf{9,25,125}$
c) $78,345 - 56,781 = \mathbf{21,564}$	f) $2,42,565 - 4,525 - 99,483 = \mathbf{1,38,557}$

2. Match the following by using the properties of subtraction:

a) $7,43,261 - 0$	--->	iii) $\mathbf{7,43,261}$ (Identity Property: Subtracting 0 doesn't change the number)
b) $7,43,261 - 1$	--->	i) $\mathbf{7,43,260}$ (Predecessor Property: Subtracting 1 gives predecessor)
c) $7,43,261 - 7,43,261$	--->	ii) $\mathbf{0}$ (Subtracting a number from itself results in 0)

3. Solve the following:

- a) What should be added to 5,65,423 to get a total of 6,00,000?

Answer: $6,00,000 - 5,65,423 = \mathbf{34,577}$

- b) What should be subtracted from 5,00,001 to get a difference of 45,000?

Answer: $5,00,001 - 45,000 = \mathbf{4,55,001}$

- c) By how much is 2,00,007 larger than 7977?

Answer: $2,00,007 - 7,977 = \mathbf{1,92,030}$

- d) What number is 2578 less than 2,98,767?

Answer: $2,98,767 - 2,578 = \mathbf{2,96,189}$

- e) What number is 8885 more than 46,578?

Answer: $46,578 + 8,885 = \mathbf{55,463}$

- f) From which number should we subtract 3750 to get 98,922?

Answer: $98,922 + 3,750 = \mathbf{1,02,672}$

Workout Word Problems (Page 43):

4. A caterer has to make arrangements for mineral water... 2175 expected in morning, 2965 in afternoon. How many bottles?

- Morning guests = 2,175
- Afternoon guests = 2,965
- Total bottles needed = $2,175 + 2,965 = \mathbf{5,140}$ bottles

5. Earth diameter is 12,742 km. Jupiter diameter is 1,39,822 km. Find difference and round to nearest Hundred.

- Difference = $1,39,822 - 12,742 = 1,27,080$ km
- Rounding to nearest Hundred = **1,27,100 km** (since Tens digit is 8, we round up)

6. State A = 38,863 sq km, State B = 1,30,058 sq km, State C = 1,91,791 sq km. Find sum of A & B, find difference between sum and C, write in words.

- Sum of State A & B = $38,863 + 1,30,058 = 1,68,921$ sq km
- Difference between State C and the sum = $1,91,791 - 1,68,921 = 22,870$ sq km
- In words: **Twenty-two thousand eight hundred seventy**

7. Ram had ₹ 15,000. Purchases mobile (₹ 6,500) and music system (₹ 5,000). How much left?

- Total spent = $6,500 + 5,000 = ₹ 11,500$
- Money left = $15,000 - 11,500 = ₹ 3,500$

8. Painter needs 10,000 cans. Has 6,587. Brought 239 from friend. How many still short?

- Total cans in hand = $6,587 + 239 = 6,826$ cans
- Short of = $10,000 - 6,826 = 3,174$ cans

9. What will be the difference if the sum of 3,64,657 and 7,573 is subtracted from 7,89,887?

- Sum = $3,64,657 + 7,573 = 3,72,230$
- Difference = $7,89,887 - 3,72,230 = 4,17,657$

10. Which is greater—sum of 10,232 and 5,987 or difference between 20,023 and 17,899? And by how much?

- Sum = $10,232 + 5,987 = 16,219$
- Difference = $20,023 - 17,899 = 2,124$
- The **Sum is greater**.
- By how much: $16,219 - 2,124 = 14,095$

Mental Maths, HOTS & Life Skills (Pages 44-46)

Mental Maths (Page 44):

- a) $56,024 - 100 = 55,924$ (Hundreds digit decreased by 1)
 b) $5,472 - 1,000 = 4,472$ (Thousands digit decreased by 1)
 c) $76,105 - 100 = 76,005$ (Hundreds digit decreased by 1)
 d) $5,67,098 - 1,000 = 5,66,098$ (Thousands digit decreased by 1)
 e) $7,80,162 - 10 = 7,80,152$ (Tens digit decreased by 1)
 f) $4,14,275 - 100 = 4,14,175$ (Hundreds digit decreased by 1)

HOTS - Higher Order Thinking Skills (Page 45):

Question: Write ten ways in which you can obtain 10,000 using addition and subtraction of numbers.

Here are 10 illustrative expressions to achieve exactly 10,000:

1. $5,000 + 5,000 = 10,000$ (Simple equal addition)	2. $12,000 - 2,000 = 10,000$ (Subtraction from larger thousand)
3. $9,999 + 1 = 10,000$ (Adding 1 to predecessor)	4. $15,000 - 5,000 = 10,000$ (Subtraction of multiple of 5000)
5. $9,500 + 500 = 10,000$ (Rounding off addition)	6. $20,000 - 10,000 = 10,000$ (Halving subtraction)
7. $8,500 + 1,500 = 10,000$ (Adding Hundreds/Thousands combo)	8. $10,500 - 500 = 10,000$ (Subtracting surplus hundreds)
9. $1,000 + 9,000 = 10,000$ (Power of ten addition)	10. $10,000 - 0 = 10,000$ (Identity property of subtraction)

Life Skills and Values (Page 46):

Question: An old age home requires ₹ 5,24,731 to meet its monthly expenses. It received a donation of ₹ 2,34,848. It also collected ₹ 1,86,748 from the sale of handicrafts. How much more money is required to meet the expenses?

Solution:

- Total monthly expense required = ₹ 5,24,731
- Amount received from donation = ₹ 2,34,848
- Amount collected from handicrafts sale = ₹ 1,86,748
- Total funds available = ₹ 2,34,848 + ₹ 1,86,748 = ₹ 4,21,596
- Additional funds required = ₹ 5,24,731 - ₹ 4,21,596 = ₹ 1,03,135

This marks the end of the Answer Key. All results have been mathematically verified with 100% precision.